《数据库系统原理》实验报告（3）					
题目：数据库安全性					
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实验环境：MariaDB

实验步骤及结果截图：

1.建 major 数据库

```
MariaDB [TeachingManagment]> CREATE DATABASE Major;
Query OK, 1 row affected (0.005 sec)

MariaDB [TeachingManagment]> show databases;
+--------------------+
| Database           |
+--------------------+
| Major              |
| TeachingManagment  |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.004 sec)
```

2.建 players 表

```
MariaDB [TeachingManagment]> CREATE TABLE Players (
    ->     Sno VARCHAR(10) PRIMARY KEY,
    ->     Sname VARCHAR(20) UNIQUE,
    ->     Ssex VARCHAR(1),
    ->     Sage SMALLINT,
    ->     Steam VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.015 sec)

MariaDB [TeachingManagment]> desc Players;
+-------+-------------+------+-----+---------+-------+
| Field | Type        | Null | Key | Default | Extra |
+-------+-------------+------+-----+---------+-------+
| Sno   | varchar(10) | NO   | PRI | NULL    |       |
| Sname | varchar(20) | YES  | UNI | NULL    |       |
| Ssex  | varchar(1)  | YES  |     | NULL    |       |
| Sage  | smallint(6) | YES  |     | NULL    |       |
| Steam | varchar(20) | YES  |     | NULL    |       |
+-------+-------------+------+-----+---------+-------+
5 rows in set (0.005 sec)
```

3.插入元组

```
MariaDB [TeachingManagment]> INSERT INTO Players (Sno, Sname, Ssex, Sage, Steam) VALUES
    -> ('200215121', 'NiKo', 'M', 27, 'G2'),
    -> ('200215122', 'm0NESY', 'M', 18, 'G2');
Query OK, 2 rows affected (0.009 sec)
Records: 2  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> select * from Players;
+-----------+--------+------+------+-------+
| Sno       | Sname  | Ssex | Sage | Steam |
+-----------+--------+------+------+-------+
| 200215121 | NiKo   | M    |   27 | G2    |
| 200215122 | m0NESY | M    |   18 | G2    |
+-----------+--------+------+------+-------+
2 rows in set (0.002 sec)
```

4.创建用户 Value 并赋予权限

```
MariaDB [TeachingManagment]> CREATE USER 'Valve'@'%' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.006 sec)
```

```
MariaDB [TeachingManagment]> GRANT CREATE ON *.* TO 'Valve'@'%' WITH GRANT OPTION;
Query OK, 0 rows affected (0.006 sec)
```

查询资料后发现，在 MariaDB 中，ALTER USER 和 DROP USER 并不总是能直接作为 GRANT 语句中的独立权限关键字。MariaDB 的权限管理逻辑通常将这些操作统一归类在 SUPER 或 CREATE USER 权限下

5.收回部分权限

```
MariaDB [TeachingManagment]> REVOKE CREATE USER ON *.* FROM 'Valve'@'%';
Query OK, 0 rows affected (0.007 sec)
```

6.在 Value 用户下查询 Players

先更换用户，再进行查询

```
# mariadb -u Valve -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 6
Server version: 12.2.2-MariaDB-ubu2404 mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> use Major;
Database changed
MariaDB [Major]> select * from players;
ERROR 1142 (42000): SELECT command denied to user 'Valve'@'localhost' for table `Major`.`players`
```

7.赋予权限

```
MariaDB [Major]> GRANT SELECT, INSERT, UPDATE ON Major.Players TO 'Valve'@'%' WITH GRANT OPTION;
Query OK, 0 rows affected (0.008 sec)

MariaDB [Major]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.003 sec)
```

8.在 Valve 下查 Players

```
# mariadb -u Valve -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8
Server version: 12.2.2-MariaDB-ubu2404 mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> use Major;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [Major]> select * from Players;
+-----------+--------+------+------+-------+
| Sno       | Sname  | Ssex | Sage | Steam |
+-----------+--------+------+------+-------+
| 200215121 | NiKo   | M    |   27 | G2    |
| 200215122 | m0NESY | M    |   18 | G2    |
+-----------+--------+------+------+-------+
2 rows in set (0.004 sec)
```

9.再插入一个元组然后查询

```
MariaDB [Major]> INSERT INTO Major.Players (Sno, Sname, Ssex, Sage, Steam)
    -> VALUES ('200215123', 'ZywOo', 'M', 23, 'Vit');
Query OK, 1 row affected (0.005 sec)

MariaDB [Major]>
MariaDB [Major]> SELECT * FROM Major.Players;
+-----------+--------+------+------+-------+
| Sno       | Sname  | Ssex | Sage | Steam |
+-----------+--------+------+------+-------+
| 200215121 | NiKo   | M    |   27 | G2    |
| 200215122 | m0NESY | M    |   18 | G2    |
| 200215123 | ZywOo  | M    |   23 | Vit   |
+-----------+--------+------+------+-------+
3 rows in set (0.002 sec)
```

10.删除 Valve 用户

```
MariaDB [(none)]> DROP USER 'Valve'@'%';
Query OK, 0 rows affected (0.015 sec)
```

```
MariaDB [mysql]> SELECT Host, User, Password FROM user;
+-----------+-------------+-------------------------------------------+
| Host      | User        | Password                                  |
+-----------+-------------+-------------------------------------------+
| localhost | mariadb.sys |                                           |
| localhost | root        | *6336BD98C0A5F666DB46B1EF474FCE1217F520A9 |
| %         | root        | *6336BD98C0A5F666DB46B1EF474FCE1217F520A9 |
| 127.0.0.1 | healthcheck | *E5E5E780426991B58177310DC67A829324A2E4CC |
| ::1       | healthcheck | *E5E5E780426991B58177310DC67A829324A2E4CC |
| localhost | healthcheck | *E5E5E780426991B58177310DC67A829324A2E4CC |
| %         | lxy         | *6336BD98C0A5F666DB46B1EF474FCE1217F520A9 |
+-----------+-------------+-------------------------------------------+
7 rows in set (0.009 sec)
```

出现的问题：

1.直接在 Valve 用户下对其赋予权限失败

解决方案：

1.切换到 root 用户即可